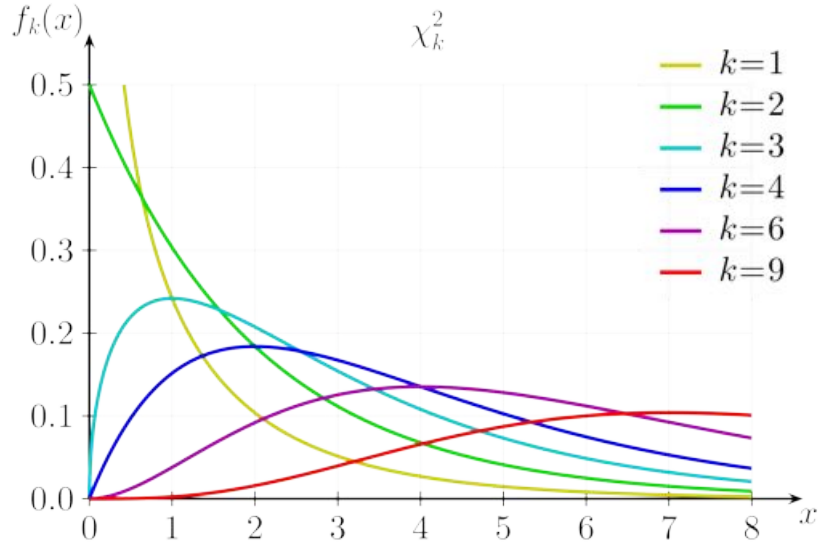
Chi-Square χ^2 Distribution

If $Z_1, \ldots, Z_k$ are independent, standard normal (Gaussian) random variables, then the sum of their squares:

$$\text{sum}_k(Z_i^2)$$

is distributed as χ^2, with "k" degrees of freedom

Mean = K
variance = 2K

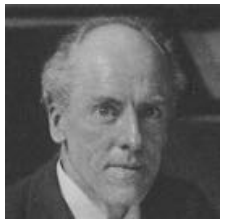


$$f(x;\, k) = \begin{cases} \dfrac{x^{\frac{k}{2}-1} e^{-\frac{x}{2}}}{2^{\frac{k}{2}} \Gamma\left(\frac{k}{2}\right)}, & x > 0; \\ 0, & \text{otherwise.} \end{cases}$$

Why is it called $\chi 2$?

Greek character Chi (χ) appears to be the notation used by Karl Pearson in 1900:

"On the Criterion that a Given System of Deviations from the Probable in the Case of a Correlated System of Variables is such that it can be Reasonably Supposed to have Arisen from Random Sampling," Philosophical Magazine, 50, (1900), 157-175.



Karl Pearson
1857 – 1936
English mathematician
and biostatistician

Where Gamma Function:

$$\Gamma(n) = (n-1)!$$

$$\Gamma(z) = \int_0^\infty x^{z-1} e^{-x}\, dx, \qquad \Re(z) > 0$$

Chi-Square χ^2 Goodness of Fit Test

https://stattrek.com/chi-square-test/goodness-of-fit.aspx

Pearson, 1900.
Suppose we have k mutually-exclusive classes
N - total number of all observations,
p_i - hypothetical probability of each class (null hypothesis).
sum(p_i) = 1
n_i - number of observations for each class (i=1,2,...k)
m_i - expected value of observations $m_i=N*p_i$
sum(m_i) = N*sum(p_i) = N = sum(n_i)

Karl Pearson has proposed that as N gets bigger, the sum below converges to χ^2

X^2 = sum((n_i-m_i)^2/m_i) --> **χ^2 distribution with (k – 1) degrees of freedom**
sum((observed – model)^2/model)
sum((observed – expected)^2/expected)

$$\sum \frac{(O_i - E_i)^2}{E_i}$$

Notation

χ2 is the lower-case Greek letter Chi
f_o is the Observed Frequency
f_e is the Expected Frequency

Chi-Square Statistic

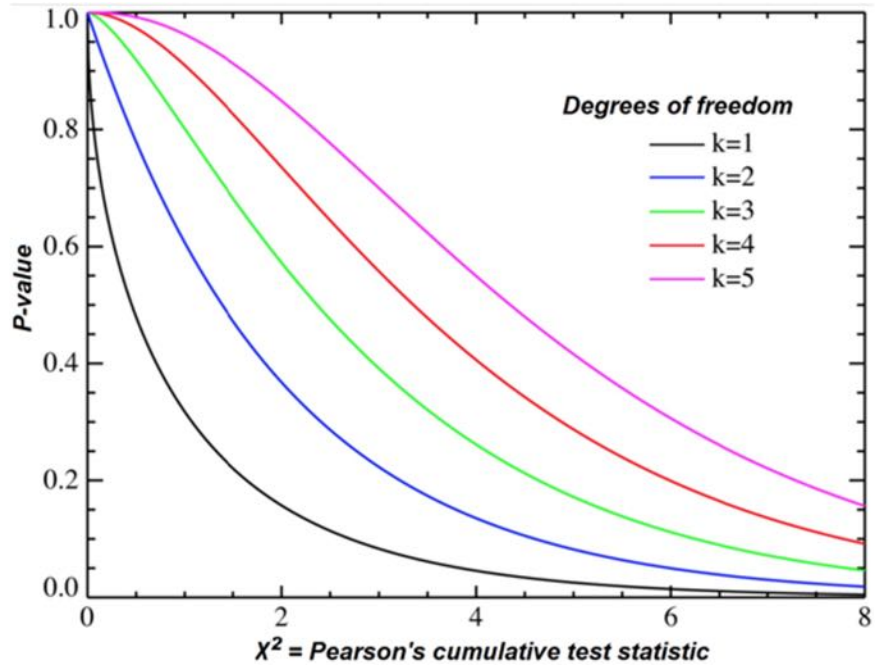
$$\chi^2 = \sum \frac{(f_o - f_e)^2}{f_e}$$

Computing Expected Frequencies

fc - frequency total for the column
fr - frequency total for the row

$$f_e = \frac{f_c f_r}{n}$$

Chi-squared test, showing χ2 on the x-axis and p-value on the y-axis.



In practice we calculate value X^2 from our data, then use χ^2 distribution to estimate the P-value (the probability that a chi-square statistic is more extreme than X^2 value that we have.

Proof and Intuition for Chi-squared test

Seven proofs of the Pearson Chi-squared independence test and its graphical interpretation – by Eric Benhamou & Valentin Melot , 2018.
- https://arxiv.org/pdf/1808.09171.pdf

Example: we have black and white balls.
The null hypothesis is that the probability of white balls is p (and black – (1-p)).
You have drawn a sample of n balls and observed k white balls.
The exact binomial probability of obtaining k white balls from n is

$P(K=k \mid n,p) = bin(n,k)p^k(1-p)^{n-k}$, where bin(n,k) is the binomial coefficient.

Now you can approximate the probability by calculating the z-score.
The mean of the binomial distribution is **nq**
and standard deviation **s = sqrt(np(1−p))**.
So the z-score is **z=(k−np)/s** and it follows standard normal distribution.

On the other hand, a squared normally distributed variable (or their sum) follows a χ2-distribution.
So, in this case the square z^2 is χ2-distributed (with 1 degree of freedom).

Note that z^2 can be expressed as $\frac{(k-nq)^2}{nq} + \frac{(n-k-n(1-q))^2}{n(1-q)}$

which is a more familiar expression for the χ2-statistic.

for white balls:
np = expected value
k = observed value
for black balls:
n(1-p) = expected value
n-k = observed value

Asymptotical proof

if k=2

$$T = \frac{(X_1 - np_1)^2}{np_1} + \frac{(X_2 - np_2)^2}{np_2}$$
$$= \frac{(X_1 - np_1)^2}{np_1} + \frac{(n - X_1 - n[1 - p_1])^2}{n(1 - p_1)}$$
$$= \frac{(X_1 - np_1)^2}{np_1} + \frac{(X_1 - np_1)^2}{n(1 - p_1)}$$
$$= \frac{(X_1 - np_1)^2}{np_1(1 - p_1)} \quad \text{since} \quad \frac{1}{p} + \frac{1}{1-p} = \frac{1}{p(1-p)}$$

X_1 is a binomial random variable with mean np_1 and variance $np_1(1 - p_1)$

$$\frac{(X_1 - np_1)}{\sqrt{np_1(1 - p_1)}} \sim N(0,1)$$

$$\left(\frac{(X_1 - np_1)}{\sqrt{np_1(1 - p_1)}}\right)^2 \sim X_{,1}^2$$

Chi-squared with 1 degree of freedom

Example of χ^2 test - goodness of fit

(from https://stattrek.com/chi-square-test/goodness-of-fit.aspx)

Problem:

We have 3 types of cards (T1,T2,T3) with theoretical frequency of (30%, 60%, 10%).
We have a random sample of 100 cards with counts (50, 45, 5).
Is this sample consistent with theory? Use a 0.05 level of significance.

Solution:

Null hypothesis: The proportions are (30%, 60%, 10%)
Alternative hypothesis: At least one of the proportions in the null hypothesis is false.
Degrees of Freedom DF = k-1 = 3-1 = 2
Expected values:
$E_1 = 100*0.3 = 30$, $E_2 = 100*0.6 = 60$, $E_3 = 100 * 0.10 = 10$
$X^2 = \Sigma [(O_i - E_i)^2 / E_i] = [(50 - 30)^2 / 30] + [(45 - 60)^2 / 60] + [(5 - 10)^2 / 10] = 19.58$

The P-value is the probability that a chi-square statistic having 2 degrees of freedom is more extreme than 19.58.
Using Chi-Square Distribution we find P(X2 > 19.58) = 0.0001.

Since the P-value (0.0001) is less than the significance level (0.05), we reject the null hypothesis.

Note: If you use this approach on an exam, you may also want to mention why this approach is appropriate.
Specifically, the approach is appropriate because the sampling method was simple random sampling, the variable under study was categorical, and each level of the categorical variable had an expected frequency count of at least 5.

Example of χ^2 test – coin flip

Problem:

We want to test a null hypothesis that the coin is fair.
We flip the coin N times and got H heads.
So number of observed tails T = N-H.

Expected values are N/2 for heads and for tails.

X2 = sum((observed – expected)^2/expected) =

$= [(H - N/2)^2 / (N/2)] + [(T - N/2)^2 / (N/2)]$

The P-value is the probability that a chi-square statistic having one degree of freedom is more extreme than that value.

Chi-Square χ^2 Test for Independence

(two-way tables, association)

Evaluates the relationship between two variables.
It is a non-parametric test, performed on categorical data.

Example:
500 boys and girls are asked which is their favorite color:
blue, green, or pink (see results).
Using alpha = 0.05, would you conclude that there is a relationship between gender and favorite color?

Degrees of Freedom df = (rows-1)(columns-1) = (2-1)(3-1) = 2

In χ^2 table the value for alpha=0.05 and df=2 is 5.99

Calculate statistics: χ^2 = sum $(f_o - f_e)^2 / f_e$ = 276.39 >> 5.99

Conclusion: yes, there is a relationship

OBSERVED	Blue	Green	Pink	
Boys	100	150	20	300
Girls	20	30	180	200
	120	180	200	N = 500

EXPECTED	Blue	Green	Pink	
Boys	72	108	120	300
Girls	48	72	80	200
	120	180	200	N = 500

Chi-Square χ^2 Test of Homogeneity

from https://courses.lumenlearning.com/wmopen-concepts-statistics/chapter/test-of-homogeneity/

	Admit Steroid Use		
	Yes	No	Totals
Division I	103	8,440	8,543
Division II	52	4,289	4,341
Division III	65	6,428	6,493
Totals	220	19,157	19,377

Null hypothesis:
steroid use doesn't differ between divisions.

Degrees of Freedom df = (rows-1)(columns-1) = (2-1)(3-1) = 2

In χ^2 table looking for value for (alpha=0.05 and df=2): χ^2 = 5.99

Calculate statistics: χ^2 = sum $(f_o - f_e)^2 / f_e$ = 1.57

The smaller is the experimental chi-square statistics – the larger is P-value.
(P-value = 0.4561 for χ^2 = 1.57)

So, as 1.57 < 5.99, we conclude that we can not reject the null hypothesis.
The data does not provide strong enough evidence to conclude that steroid use differs in the three divisions (P-value = 0.4561).

Expected Counts			
	Admit Steroid Use		
	Yes	No	Totals
Division I	96.96	8446.04	8,543
Division II	49.27	4291.73	4,341
Division III	73.70	6419.30	6,493
Totals	220	19,157	19,377

Expected percent is 220/19,377 = 1.135%
1.135% of 8,543

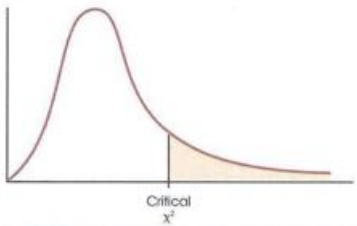
Expected percent is 220/19,377 = 1.135%
1.135% of 4,341

Expected percent is 220/19,377 = 1.135%
1.135% of 6,493

steroid use "yes" *steroid use "no"*

Division I *Division II* *Division III* *Division I* *Division II* *Division III*

$$\chi^2 = \frac{(130-96.96)^2}{96.96} + \frac{(52-49.27)^2}{49.27} + \frac{(65-73.70)^2}{73.70} + \frac{(8440-8446.04)^2}{8446.04} + \frac{(4289-4291.73)^2}{4291.73} + \frac{(6428-6419.30)^2}{6419.30} \approx 1.57$$



χ^2

	Proportion in Critical Region				
df	0.10	0.05	0.025	0.01	0.005
1	2.71	3.84	5.02	6.63	7.88
2	4.61	5.99	7.38	9.21	10.60
3	6.25	7.81	9.35	11.34	12.84
4	7.78	9.49	11.14	13.28	14.86
5	9.24	11.07	12.83	15.09	16.75
6	10.64	12.59	14.45	16.81	18.55
7	12.02	14.07	16.01	18.48	20.28
8	13.36	15.51	17.53	20.09	21.96
9	14.68	16.92	19.02	21.67	23.59
10	15.99	18.31	20.48	23.21	25.19
11	17.28	19.68	21.92	24.72	26.76
12	18.55	21.03	23.34	26.22	28.30
13	19.81	22.36	24.74	27.69	29.82
14	21.06	23.68	26.12	29.14	31.32
15	22.31	25.00	27.49	30.58	32.80
16	23.54	26.30	28.85	32.00	34.27
17	24.77	27.59	30.19	33.41	35.72
18	25.99	28.87	31.53	34.81	37.16
19	27.20	30.14	32.85	36.19	38.58
20	28.41	31.41	34.17	37.57	40.00
21	29.62	32.67	35.48	38.93	41.40
22	30.81	33.92	36.78	40.29	42.80
23	32.01	35.17	38.08	41.64	44.18
24	33.20	36.42	39.36	42.98	45.56
25	34.38	37.65	40.65	44.31	46.93
26	35.56	38.89	41.92	45.64	48.29
27	36.74	40.11	43.19	46.96	49.64
28	37.92	41.34	44.46	48.28	50.99
29	39.09	42.56	45.72	49.59	52.34
30	40.26	43.77	46.98	50.89	53.67
40	51.81	55.76	59.34	63.69	66.77
50	63.17	67.50	71.42	76.15	79.49
60	74.40	79.08	83.30	88.38	91.95
70	85.53	90.53	95.02	100.42	104.22
80	96.58	101.88	106.63	112.33	116.32
90	107.56	113.14	118.14	124.12	128.30
100	118.50	124.34	129.56	135.81	140.17